Emergent Behavioural Signalling in Cooperative Multi-Agent Highway Driving

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***Abstract*—Autonomous highway driving is a multi-agent coordination problem in which every vehicle carries a private behavioural type that others cannot observe directly. This thesis formulates cooperative multi-agent highway driving as a repeated signalling game over a small discrete vocabulary of behaviourally grounded tokens whose meanings are not pre-assigned. It then asks whether communication emerges among the agents at all, and in what sense. The agents are trained using multi-agent reinforcement learning and the signalling channel is probed with evaluation-time interventions that separately disturb the token, the neighbour kinematics, and their binding.**

Communication does emerge, not in the form standard information metrics are designed to register, but through what we call *convention-of-use*. The token carries almost no information about a driver's hidden type. Nevertheless, corrupting the binding between token and behaviour raises the collision rate by 65 to 86%, depending on how the binding is broken. In this experiment, the active interventions revealed a dependence on the channel that the passive metrics did not register. This suggests that evaluations of emergent communication, and audits of connected-driving systems, should test the binding between message and behaviour directly rather than rely on measured informativeness alone.

***Index Terms*—multi-agent reinforcement learning, emergent communication, signalling games, autonomous driving, convention**

I. Introduction

Autonomous vehicles have gradually moved from test tracks onto public roads. Robotaxis have been carrying passengers in several cities, and driver-assistance systems have been deployed in commercial cars. More and more manufacturers are building these systems, and no two are alike. They run different sensors, different planners, and different driving policies, trained separately and opaque to one another. For a long time to come they will also share the road with human drivers rather than replace them, so any one car has to cope with mixed, heterogeneous traffic it neither controls nor can see inside. This makes how such vehicles communicate with each other important. But whether communication can emerge among these vehicles is still unsettled. This paper asks that question, in a small, controlled version of the problem.

Driving on a highway means constantly reacting to cars you cannot fully read. Whether a neighbour is cautious or assertive, how much risk they accept, how close they like to follow, none of this is visible directly. The usual approach in the field is perception, prediction, and planning: build a model of how nearby cars are likely to move, and plan around it [1]. These prediction methods have improved a great deal, from simple rule-based simulation, to tracking a few possible intentions [2, 3], to learning a hidden driver type from data with neural-network encoders [4–6].

But this whole line of work looks at only one side of the problem. The car asks "what will my neighbour do?" and stops there. A human driver also asks the other half: "what do our neighbours think I am, and is the way I drive shaping that belief?" How a driver acts matters as much as how it reads and responds to others, since that action is what lets other drivers form a belief about its type. Those interactions act as signals that help agents coordinate with each other [7–9].

Take a simple two-car example. Car A is cautious and car B is assertive, and neither can see the other's type directly. The two approach a merge where only one can go first. B wants A to know that it will not back off, so it makes that intent legible in the only way available: it closes the gap quickly, holds a tight headway, and accelerates decisively towards the merge point. None of this is a message in any literal sense. It is simply how B drives, but executed firmly enough that the style itself announces an assertive type.

A reads that style, infers an assertive neighbour, and answers with its own: it eases off, opens the gap, and lets B take the space. That cautious response is itself legible to B, which now sees a driver willing to give way. The two settle without a collision and without any negotiation, as B goes first and A follows. Neither changed who it is. B stayed assertive and A stayed cautious; they did not meet in the middle on a shared style, but read each other's style and each kept its own, and that mutual reading let them coordinate. Neither car sent a message. The coordination came entirely from the visible manner of ordinary moves, where the same manoeuvre carries different information depending on *how* it is executed.

This is the setting studied by signalling-game theory [10–12]. An informed sender gives off a signal; an uninformed receiver reads it and updates what it believes. Outcomes fall into three cases: separating, where each type sends a different signal and the signal is fully informative; pooling, where every type sends the same signal and it tells you nothing; and partial-pooling in between [13].

This paper treats cooperative multi-agent highway driving as a signalling game and builds tools to determine which case actually appears when agents learn to use a shared set of discrete tokens whose meanings are left open. To keep every token tied to real behaviour, the signalling channel is grounded: sending a token forces the sender into a matching driving style. Moreover, we keep the agents cooperative on purpose. That way, if they end up pooling, it cannot be because someone is lying for an advantage, so the result tells us about

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the structure of the game rather than about clever deception.

By "cooperative" we do not mean a common-payoff game. Each agent maximises its own return, weighted by its type, and the two types value speed differently, so payoffs are general-sum. We use the term in a narrower sense, with three parts. First, interests are aligned on the outcome that dominates the reward: every reward term depends only on the agent's own collision, speed, lane and road state, a collision penalises every vehicle involved, and no agent is rewarded for another's loss. Second, there is no incentive to deceive, since nothing rewards an agent for what others believe about its type. Third, all agents share one set of policy parameters and are trained jointly with a centralised critic, so the population learns as a team rather than as competing learners. What remains in conflict is who yields and who takes a gap; Section V treats this residual conflict as the mixed-motive part of a coordination-dominant game.

Framing it this way also changes how we should judge success. Work on emergent communication in multi-agent systems (MAS) asks exactly this question of how agents learn to use a shared set of signals [14–16]. It usually measures success with information: a channel counts as communication if its messages tell you something about the sender's hidden state. That test assumes a signal exists to reveal private information [10, 11], and it is normally read off a trained model without touching it. But a message can align with a hidden state without the receiver ever using it [17, 18], and a channel whose real job is to coordinate behaviour, not to reveal type, can be invisible to such a test. In a safety-critical setting, a metric that cannot tell a useless channel from an important one is dangerous.

Concretely, we ask two research questions:

- **RQ1**: Does type-contingent signalling emerge at all, when cooperative agents share a grounded set of tokens whose meanings are left open?
- **RQ2**: Is any such emergence driven by type-dependent reward heterogeneity between the drivers, rather than by the presence of the signalling architecture itself?

We train groups of six agents with Multi-Agent Proximal Policy Optimisation (MAPPO) [19] under four conditions, twenty seeds each, each isolating one factor. The *bimodal* is the full model, with heterogeneous driver types and an intact channel. The *same-reward* keeps the channel but gives both types identical rewards, removing the type-dependent reward heterogeneity. The *random-signal* keeps that reward heterogeneity but replaces the emitted token with a random draw, severing type from signal, giving a baseline that any real emergence has to beat. The *no-signal* removes the channel entirely, leaving only neighbour kinematics, as a plain kinematics-only baseline.

We run a set of interventions at evaluation time that separately disturb the token, the neighbour kinematics, and the link between them. We also introduce the Counterfactual Decoding Effect (CDE), a receiver-side measure that reads how much a receiver's action changes, in real control units, when only the token it receives is swapped. We report it alongside the information-based Causal Influence of Communication (CIC) [17]. Finally we read the results through the difference between signals that reveal private information and conventions that only coordinate.

This paper makes four contributions. First, it formally frames cooperative highway driving as a learned signalling game over a discrete, behaviour-grounded set of tokens with open meanings. It studies the emergent convention itself rather than only the driving policy it produces. Second, it shows that the channel carries almost no type information but that its integrity still matters. Breaking the link between token and behaviour raises the collision rate by 65% to 86%, depending on how the link is broken. At the same time, a population trained with no channel at all shows no detectable difference in collision rate or return. Third, it introduces the Counterfactual Decoding Effect, an action-space measure of decoding that agrees closely with the information-based measure across seeds. Fourth, it offers a caution for the method, drawn from what happened here rather than asserted in general. In this experiment, the active interventions exposed a dependence on the channel that the passive metrics did not register. This suggests that checking whether communication really emerged, and auditing connected-driving systems, should test the binding between message and behaviour directly rather than rely on measured informativeness alone.

This paper is structured as follows. Section II highlights prominent studies in the literature and positions our approach within this body of work. Section III sets out the formulation and the evaluation protocol, Section IV presents these findings, and Section V reads them through the game theory of signalling and convention.

II. Related Work

Signalling games. In a signalling game an informed sender observes a private type and emits a costless message; an uninformed receiver updates beliefs via Bayes' rule and responds accordingly [10]. Cheap-talk channels admit a continuum of equilibria from fully separating to pooling, and [11] showed that signal costs break this indeterminacy: when types bear different costs for the same signal, honest separation becomes optimal. This motivates the execution-profile signal space used here, where each token trades off differently against the reward components of cautious versus assertive types, giving structural cost asymmetry without pre-assigned semantics. [13] supplies the Perfect Bayesian Equilibrium solution concept, used here only as a diagnostic taxonomy (separating/partial-pooling/pooling) for characterising what training produces, not as an analytical target.

Game-theoretic lane-changing. A body of work applies signalling and game theory directly to lane-changing: [20] solved discretionary lane-changing as a signalling game analytically under explicit payoffs; [21] and [22] modelled mandatory lane changes as strategic games in connected-vehicle settings; [23] built a game-theoretic controller estimating merge-opponent aggressiveness; [7] analysed pedestrian crossings as a chicken game, and [8] documented how road users read vehicle movement as implicit type signals. These works solve analytically for equilibria under assumed

payoffs or hand-specify the signal-behaviour mapping. Multi-agent reinforcement learning for cooperative driving does learn coordination among vehicles, but it treats communication as an engineered channel, either sharing observations over a given communication topology [24] or learning to transmit encoded state and intent [25], and evaluates it by task performance rather than by what the messages come to mean. To our knowledge, no prior work trains driving agents to discover whether a useful convention emerges from uninterpreted interaction over a vocabulary whose semantics are left open; this is the gap this paper addresses.

Behavioural type inference. The standard approach to heterogeneity is passive inference: [2] used an intention-aware POMDP marginalising over intent hypotheses; [3] estimated multi-policy behaviour distributions for trajectory planning; [6] coupled latent-type inference to the RL objective; [4] and [5] trained unsupervised driver-trait encoders that improve collision avoidance when conditioned on. All frame the problem one-sidedly, asking only what a neighbour is rather than what the neighbour infers about the ego vehicle. [9] showed human driving already carries type-revealing kinematic structure absent explicit signalling, and [8] documented that humans actively read and produce movement gestures to coordinate – suggesting the substrate for emergent signalling already exists in ordinary driving, which this paper asks whether MARL agents exploit deliberately.

Emergent communication in MAS. [14] showed agents learn a discrete channel end-to-end via differentiable inter-agent communication; [15] showed continuous message passing supports coordination without a hand-designed protocol; [16] surveyed a decade of referential-game results showing agents reliably develop shared symbol-meaning mappings when the channel is informative for reward. Two methodological results shape the evaluation design here: [18] showed agents frequently converge to pooling unless training explicitly encourages positive signalling (diverse messages) and positive listening (behaviour that responds to messages), and [17] formalised these as Speaker Consistency and Causal Influence of Communication, the metrics adopted below.

III. Methodology

A. Problem formulation

The highway scenario is a Partially Observable Stochastic Game $\mathcal{G} = (\mathcal{I}, \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{M}, T, R, \gamma)$, where $\mathcal{I} = \{1, \dots, N\}$ is the set of controlled agents, $\mathcal{S}$ is the joint environment state, $\mathcal{O}$ is the observation space, $\mathcal{A} = [-1, 1]^2$ is a continuous action (longitudinal acceleration, steering), $\mathcal{M} = \{0, 1, 2\}$ is a discrete signal token, T is the transition function, R is the reward function, and $\gamma \in (0, 1)$ is the discount factor.

Each agent $i \in \mathcal{I}$ carries a latent type $\theta_i \in \Theta = \{0, 1\}$ representing cautious and assertive styles of driving, respectively. It is drawn independently at episode reset from a fixed population distribution p_Θ. In the bimodal condition, $p_\Theta(0) = p_\Theta(1) = 0.5$. Types are private and persist for the full episode, and there are no persistent agent identifiers across episodes.

At each decision step t, agent i selects $(a_i^t, m_i^t) \in \mathcal{A} \times \mathcal{M}$. The token m_i^t is bound to an execution profile $\pi(m_i^t)$ that modifies the kinematic envelope within which a_i^t is rendered; more detail on the execution profiles is in Section III-D.

The policy for agent i decomposes into two heads sharing an encoder. The signal head commits to a token first, and the response head then conditions on that committed token:

$$\pi_i(a_i^t, m_i^t \mid o_i^t, \theta_i) = \underbrace{\sigma_i(m_i^t \mid o_i^t, \theta_i)}_{\text{signal head}} \times \underbrace{\pi_i^{\text{drive}}(a_i^t \mid o_i^t, m_i^t)}_{\text{response head}}. \quad (1)$$

The signalling strategy $\sigma_i : \mathcal{O} \times \Theta \to \Delta(\mathcal{M})$ maps local observation and own type to a distribution over tokens. The response strategy $\pi_i^{\text{drive}} : \mathcal{O} \times \mathcal{M} \to \Delta(\mathcal{A})$ maps local observation and the committed signal token to a continuous action distribution. Conditioning on m_i^t lets the response head specialise to the committed kinematic envelope. The own type θ_i is not an input to the response head, so any type-consistent driving must pass through the token.

Agent i maintains an implicit belief $b_i(\theta_j \mid h_j)$ over the type of each visible neighbour j, where h_j is the observed behavioural history. The analytical reference point is on-path Bayesian updating:

$$b_i(\theta_j \mid m_j^t, h_j) \propto \sigma_j(m_j^t \mid s_j, \theta_j)\, b_i(\theta_j \mid h_j). \quad (2)$$

In this work, b_i is not maintained explicitly. Instead, the type encoder (Section III-F) produces a distributed representation of b_i from the raw observation window, with gradients flowing from the downstream loss.

B. Simulation environment

The simulation environment is a multi-agent extension of the gym-based `highway-env` simulator [26]. The road is a straight three-lane highway. Each episode runs for 40 seconds at a control frequency of 5 Hz, yielding 200 decision steps at a decision interval of $\Delta t = 0.2$ s.

All $N = 6$ vehicles on the road are policy-controlled learning agents, and no background traffic is added. The study focuses solely on the behaviour that emerges among the controlled agents, and the environment is kept as simple as possible so that scripted human-driven traffic cannot introduce additional dynamics that would confound the signal.

C. Observation space

At each step t, agent i observes its own state plus a short history for each of its $K = 5$ nearest neighbours:

$$o_i^t = \left(o_i^{\text{self}}, \{O_{ij}, M_{ij}\}_{j=1}^{K}\right), \quad (3)$$

where $o_i^{\text{self}} \in \mathbb{R}^5$ is the agent's own normalised kinematic state (presence, longitudinal and lateral position, longitudinal and lateral velocity).

For each neighbour j, $O_{ij} \in \mathbb{R}^{W \times 5}$ is a W-step history of five per-step kinematic features: whether the slot is occupied (presence), the longitudinal and lateral offset to the neighbour (Δx, Δy), and the neighbour's longitudinal and lateral velocity relative to agent i (Δv_x, Δv_y). The window length equals the neighbour count, $W = K = 5$, so each observation spans the last second of interaction. $M_{ij} \in \{-1, 0, 1, 2\}^W$ is the matching W-step history of tokens emitted by neighbour j. When fewer than K neighbours are within range, the unused slots are padded with -1.

TABLE I: Signal token to kinematic execution profile mapping.

Token	Profile	Headway mult.	Jerk cap (m/s^3)	Gap accept (m)	Lateral drift
0	Conservative	1.5	1.0	15.0	0.3
1	Neutral	1.0	2.0	10.0	0.6
2	Assertive	0.7	4.0	6.0	1.0

D. Execution profiles

Token $m \in \{0, 1, 2\}$ maps to a kinematic profile $\pi(m)$ (Table I). This mapping is fixed by design, identical across all agents and seeds. Emitting a token deterministically selects the corresponding kinematic profile, whatever the sender's private type. What is learned is not this binding but two other things: (1) which token a sender chooses to emit (the signalling strategy σ_i), and (2) how a receiver's driving responds to a received token (the response strategy π_i^{drive}).

Four quantities parameterise each profile. The headway multiplier scales the desired following distance relative to a neutral baseline, so values above 1 increase the target gap and values below 1 reduce it. The jerk cap bounds the per-step change in acceleration, controlling smoothness of longitudinal dynamics. The gap accept sets the base threshold distance to the vehicle ahead below which forward acceleration is suppressed; it is multiplied by the headway multiplier to give the effective desired gap g^*. The lateral drift amplifies steering commands whose magnitude exceeds a small threshold (0.1); higher values produce more decisive lane changes observable in the kinematic history.

The longitudinal acceleration is jerk-capped each step:

$$a_t = \text{clip}\big(\tilde{a}_t,\ a_{t-1} - \kappa\Delta t,\ a_{t-1} + \kappa\Delta t\big), \tag{4}$$

where $\tilde{a}_t$ is the raw policy output, a_{t-1} is the previous shaped acceleration, κ is the profile jerk cap, and $\Delta t = 0.2\,\text{s}$. The cap enforces physically plausible acceleration trajectories; without it, the policy could produce instantaneous acceleration reversals that real vehicles cannot execute. When a front vehicle is within the desired gap $g^* = \text{gap_accept} \times \text{headway_mult}$, positive acceleration is scaled by g/g^*, where g is the current bumper-to-bumper gap.

E. Reward structure

The per-agent reward is a standard driving reward over collision avoidance, forward progress, lane preference, and on-road presence, weighted according to the agent's private type θ_i. The weight vectors per condition are shown in Table II. For agent i at step t,

$$r_i^t = w_{\theta_i}^{col} c_i^{col} + w_{\theta_i}^{spd} c_i^{spd} + w_{\theta_i}^{lane} c_i^{lane} + w_{\theta_i}^{road} c_i^{road}, \tag{5}$$

where the four components are computed from the agent's own state alone:

$$c_i^{col} = \mathbb{1}[\text{crashed}], \quad c_i^{spd} = \text{clip}\Big(\tfrac{v_i - v_{\min}}{v_{\max} - v_{\min}}, 0, 1\Big), \tag{6}$$

$$c_i^{lane} = \tfrac{\ell_i}{L-1}, \quad c_i^{road} = \mathbb{1}[\text{on road}]. \tag{7}$$

c_i^{col} is the crash indicator for the current step. c_i^{spd} is agent i's speed normalised onto the reward speed range $[v_{\min}, v_{\max}] = [20, 30]$ m/s and clipped, so it saturates at 1 rather than paying for unbounded speed. c_i^{lane} indexes the lane occupied, $\ell_i \in \{0, \ldots, L-1\}$ with $L = 3$, normalised so the leftmost lane scores 0 and the rightmost 1; this is a right-lane preference in the convention of the base simulator [26], not a lane-keeping penalty. c_i^{road} indicates agent i is still on the drivable surface. All four are bounded in $[0, 1]$, so the weights in Table II are directly comparable in scale.

TABLE II: Reward weight vectors per type.

Type	w^{col}	w^{spd}	w^{lane}	w^{road}
0 (cautious)	-2.0	0.2	0.2	1.0
1 (assertive)	-1.5	0.8	0.05	1.0

The two types differ only in these weights. The cautious type pays more for a collision (-2.0 against -1.5) and is rewarded far less for speed (0.2 against 0.8), which is what makes type behaviourally legible in the kinematics and is the type-dependent reward heterogeneity that RQ2 removes.

There is no reward term for the signal, which means no bonus for type-consistent emission and no penalty for type-inconsistent emission. This is a deliberate methodological commitment. If the reward directly incentivised truthful signalling, RQ1 would be answered by construction, such that separation would emerge because we paid for it. Keeping the reward free of any signal-dependent term ensures that the regime observed is an endogenous consequence of interaction structure, not reward engineering.

F. Network architecture

Figure 1 gives an overview of the full per-agent pipeline including all components used. Each component is described in turn below.

1) TypeEncoder: A single-layer GRU [27] with hidden size $d = 64$ maps the W-step sequence to a final hidden state, which is the neighbour embedding. For each neighbour $j \in \mathcal{N}_i$, the TypeEncoder maps the W-step observation window to a fixed-size embedding. The input at window step $w \in \{0, \ldots, W-1\}$ is

$$x_{ij}^w = O_{ij}^w \,\|\, \text{one-hot}(M_{ij}^w), \tag{8}$$

where O_{ij}^w contains the five per-step kinematic features described above, one-hot(M_{ij}^w) is the standard one-hot encoding of the token, and $\|$ denotes concatenation. Recurrent encoders over temporal trajectory windows have been used for driver style inference in prior DRL work [5]. GRU is chosen over LSTM because the short window ($W = 5$) does not require LSTM's additional cell state to avoid vanishing gradients; despite this simplicity, it achieves comparable sequence modelling with fewer parameters [27]. Self-attention is also not used, as the quadratic cost and positional-encoding overhead are unnecessary for sequences of length five.

2) DeepSets aggregator: The K neighbour embeddings are aggregated with a permutation-invariant DeepSets layer [28]:

$$z_i = \rho\Big(\sum_{j \in \mathcal{N}_i} \phi(\hat{z}_{ij})\Big), \tag{9}$$

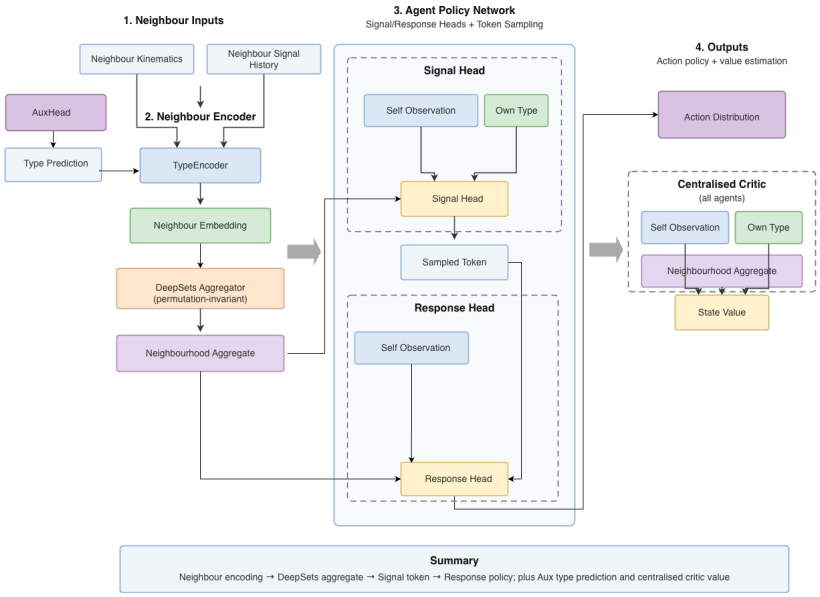


Fig. 1: System architecture diagram. Neighbour kinematics and signal history pass through the shared TypeEncoder to produce a neighbour embedding, which the permutation-invariant DeepSets aggregator reduces to a neighbourhood aggregate. The Signal Head conditions on self-observation and own type to sample a token; the Response Head then conditions on self-observation, the neighbourhood aggregate, and that sampled token to produce the action distribution. The AuxHead supervises the neighbour embedding with a privileged type-prediction loss during training only, and is discarded at evaluation. The Centralised Critic pools all agents' self-observations, types, and neighbourhood aggregates to estimate the joint state value.

where $\phi : \mathbb{R}^{64} \to \mathbb{R}^{64}$ is an MLP applied independently to each neighbour embedding and $\rho : \mathbb{R}^{64} \to \mathbb{R}^{64}$ is an MLP applied to the masked sum. Absent neighbour slots are zeroed before summation. A permutation-invariant aggregator is required in this setting: an agent's neighbours form an unordered, variable set, the slot in which a neighbour appears is arbitrary, and neighbours enter and leave as vehicles move, yet the policy must return the same decision under any reordering of the same set. Each neighbour moreover carries both its kinematics and its emitted token, and these pairs must be pooled symmetrically so that what a neighbour signals is read independently of where it happens to sit in the observation. DeepSets guarantees this invariance by construction. DeepSets [28] is preferred over Social Attention [29] because the two architectures solve different problems: Social Attention routes selectively toward high-relevance neighbours via ego-as-query, neighbour-as-key/value attention, useful when the policy must identify a specific priority vehicle. The task here instead requires estimating the neighbourhood type distribution, a summary statistic for which uniform sum pooling is provably sufficient, and DeepSets requires fewer parameters and less computation than the alternative.

3) Signal head: The signal head maps the concatenation of self-observation, own type, and neighbourhood aggregate to a categorical distribution over tokens:

$$\sigma_i(\cdot \mid o_i^{\text{self}}, \theta_i) = \text{Softmax}\big(f_\sigma([o_i^{\text{self}} \,\|\, e_\theta(\theta_i) \,\|\, z_i])\big) \in \Delta(\mathcal{M}), \tag{10}$$

where $e_\theta : \Theta \to \mathbb{R}^{16}$ is a learned type embedding and f_σ is an MLP. The token is sampled as $m_i^t \sim \sigma_i$.

4) Response head: The response head maps self-observation, neighbourhood aggregate, and the committed signal token to a bivariate action distribution:

$$\mu_i^a = f_\pi\big([o_i^{\text{self}} \,\|\, z_i \,\|\, e(m_i^t)]\big), \tag{11}$$

where $e : \mathcal{M} \to \mathbb{R}^8$ is a learned token embedding and f_π is an MLP. The token embedding allows the head to specialise its action distribution to each kinematic envelope without pre-assigning signal semantics.

5) Auxiliary type prediction head: During training, an auxiliary head attached to the TypeEncoder predicts the ground-truth type of each observed neighbour from its embedding:

$$\hat{p}(\theta_j \mid \hat{z}_{ij}) = \text{Softmax}(f_{\text{aux}}(\hat{z}_{ij})), \tag{12}$$

where f_{aux} is an MLP with a two-class output. The training cross-entropy loss is

$$\mathcal{L}_{\text{aux}} = -\frac{1}{\sum_{i,j} \mathbb{1}[\theta_j \geq 0]} \sum_{i \in \mathcal{I}} \sum_{\substack{j \in \mathcal{N}_i \\ \theta_j \geq 0}} \log \hat{p}(\theta_j \mid \hat{z}_{ij}), \tag{13}$$

where the mask $\theta_j \geq 0$ excludes unoccupied neighbour slots (type $= -1$). The head is removed at evaluation; only the trained encoder weights are retained. The auxiliary objective serves two roles. First, the PPO [30] reward (collision penalty and speed) carries no direct gradient about why a neighbour behaved a certain way; without explicit supervision the aggregate could encode proximity and velocity while ignoring latent type entirely. Dense cross-entropy supervision provides a direct gradient path into the TypeEncoder and DeepSets aggregator, forcing the shared representation to become type-discriminative. Second, signalling problems often

face a chicken-and-egg dilemma: the sender has no incentive to emit type-consistent tokens until the receiver can decode them, and the receiver representation does not encode type until the sender does so consistently [18]. The auxiliary head breaks this deadlock by supervising the receiver-side representation from raw kinematics before any convention exists, mirroring the role-inference auxiliary tasks used in cooperative MARL [31, 32]. This loss acts only on the receiver-side encoder and never touches the sender's signal head, so it does not prescribe which token a given type should emit; it makes type easier to represent from observed kinematics and past tokens.

6) Centralised critic: Under decentralised execution, each agent's local value function faces a non-stationarity problem [33]: as all agents' policies co-evolve, the transitions observed by agent i appear non-Markovian. Conditioning the critic on the joint state restores stationarity during training, since the joint state is Markovian even as individual policies change. This centralised-training decentralised-execution (CTDE) principle was introduced for multi-agent actor-critic by [33] and its effectiveness for cooperative tasks confirmed empirically by [19]. Joint state access also reduces advantage variance relative to local critics, since the value estimate is not confounded by unobserved co-agent behaviour. The centralised critic takes the joint state of all N agents:

$$V_\phi = f_V\big([o_1^{\text{self}} \,\|\, e_V(\theta_1) \,\|\, z_1 \,\|\, \cdots \,\|\, o_N^{\text{self}} \,\|\, e_V(\theta_N) \,\|\, z_N]\big) \in \mathbb{R}, \tag{14}$$

where $e_V : \Theta \to \mathbb{R}^{16}$ is a critic-private learned type embedding (separate from the policy's e_θ) and f_V is an MLP. The aggregate vectors z_i are computed with the shared policy encoder and treated as constants when updating the critic. The critic is used only during training.

G. Training procedure

We train the agents with MAPPO [19]. Training runs in iterations. Each iteration first collects $T = 512$ environment steps. It then estimates how much better than average each action was, using generalised advantage estimation (GAE) [34] normalised to zero mean and unit variance. Finally, it updates the policy (minibatch size 128, clip ratio $\epsilon = 0.2$).

The policy and the critic are trained with separate Adam optimisers [35]. Keeping them separate isolates their gradients: the policy update changes only policy parameters and the critic update changes only critic parameters, so the critic's learning signal cannot leak into the policy's representation.

To stop the agents from settling on one token too early, we add a signal-entropy bonus to the policy loss. The signal entropy is the Shannon entropy [36] of the token distribution the signal head outputs: it is near zero when an agent almost always emits the same token, and maximal when it spreads evenly over all tokens. Rewarding higher entropy therefore discourages an early collapse to a pooling equilibrium in which the token carries no information about type. The strength of this bonus is reduced linearly from $c_m^0 = 0.08$ to $c_m^{\text{end}} = 0.005$ over the first $T_{\text{anneal}} = 1200$ iterations (of 1600 total). Starting high pushes the policy to try all three tokens early, and lowering it later lets a stable convention settle once that exploration is complete. The bonus rewards spread across tokens, not any particular token-to-type pairing, so like the auxiliary head it discourages premature collapse without prescribing what the population should converge on.

H. Evaluation metrics

1) Experiment controls: We conducted experiments under four conditions, each trained with twenty independent seeds. The principal condition, *bimodal*, assigned the two driver types distinct reward weights as in Table II. The *same-reward* condition retained the full architecture but assigned both types identical weights (those of the cautious driver), removing the type-dependent reward heterogeneity while leaving the channel intact. The *random-signal* condition retained that reward heterogeneity but replaced the emitted token with a uniform draw during training, severing the type-to-token link while keeping the channel present. The *no-signal* condition retained the bimodal reward heterogeneity but removed the communication channel altogether: no token is emitted or received and agents observe only neighbour kinematics, as in a standard multi-agent driving setup, with every agent following the neutral dynamics profile since no token remains to ground one. Because it carries no token, the signalling metrics (ρ, SC, CIC, CDE) do not apply to the *no-signal* condition.

These four controls are each needed for a different reason. The *bimodal* condition is the baseline. The *same-reward* and *random-signal* conditions keep the channel intact; each isolates one factor, and together they rule out two alternative explanations for a positive ρ in *bimodal*. The *same-reward* condition rules out the possibility that any signalling-shaped architecture would show apparent signal even without reward heterogeneity to explain: if ρ in the *bimodal* condition is genuinely driven by having two types to distinguish (RQ2), removing only the reward-weight difference between types should collapse it. The *random-signal* condition rules out the possibility that the measured ρ is an artefact of the architecture rather than something earned by learning; breaking token-to-type binding, it gives the noise floor any real signal has to beat. The *no-signal* condition answers a coarser question, whether the channel matters for task outcomes at all. This condition is not a clean channel-only ablation: because the channel is grounded, profile selection has no input besides the token, so we cannot remove the token while leaving profile choice intact unless we create a new system. Its mechanism is still comparable to without-channel training in the sense that we choose a default action profile for everyone. This comparability also holds at the level of what is trained: in the *no-signal* condition, the signal head and the response head's signal-embedding table are never invoked in the forward pass and so receive no gradient, while every component that produces driving behaviour – the type encoder, the DeepSets aggregator, the response head's driving trunk, the critic, and the auxiliary type-prediction head – is optimised under the same objective as in the *bimodal* condition.

2) Task outcomes: We report two task outcomes, one for safety and one for efficiency. The collision rate is the fraction of a seed's evaluation episodes in which at least one collision occurs, reported as a percentage; lower is safer. The mean

return is the average reward an agent earns per episode, which is high when it makes steady progress at the target speed without crashing; higher is better. We compute each outcome per seed, averaging over that seed's evaluation episodes, and then summarise across the twenty seeds.

3) Type-signal mutual information: The central measurement is the empirical mutual information [36] between the latent type and the emitted signal, estimated from evaluation rollouts:

$$I(\Theta;\mathcal{M}) = \sum_{\theta\in\Theta}\sum_{m\in\mathcal{M}} p(\theta,m)\log\frac{p(\theta,m)}{p(\theta)p(m)}, \qquad (15)$$

where $p(\theta, m)$ is the empirical joint probability of type θ emitting signal m, $p(\theta)$ is the marginal probability of type θ, and $p(m)$ is the marginal probability of signal m. We report $I(\Theta;\mathcal{M})$ in bits and normalise by $\log|\Theta|$ to give a separation coefficient $\rho \in [0, 1]$. Operationally, $\rho < 0.1$ indicates *pooling* (the signal carries essentially no information about type); $\rho > 0.9$ indicates *separating* (the signal determines type); intermediate values indicate *partial-pooling*. The distribution of ρ across seeds and conditions directly addresses RQ1 and RQ2.

4) Emergent communication metrics: Speaker Consistency (SC) [37] measures how consistently each type commits to a preferred token:

$$\mathrm{SC} = \frac{1}{|\Theta|}\sum_{\theta}\max_{m\in\mathcal{M}} p(m \mid \theta). \qquad (16)$$

High SC is necessary but not sufficient for separation: a policy could achieve $\mathrm{SC} = 1$ by always emitting the same token regardless of type (pooling equilibrium). SC and ρ together distinguish pooling-with-consistency from genuine type-signal separation.

Causal Influence of Communication (CIC) [17] measures whether receivers actually change their actions in response to the received token:

$$\mathrm{CIC} = D_{KL}\Big(\pi^{\text{drive}}(\cdot \mid o,m)\,\Big\|\,\mathbb{E}_{m'\sim p(m)}\big[\pi^{\text{drive}}(\cdot \mid o,m')\big]\Big). \qquad (17)$$

CIC is estimated by sampling action distributions under actual received signals versus shuffled signals drawn from the empirical marginal $p(m)$. A near-zero CIC indicates the response head ignores the token channel, while elevated CIC confirms receivers condition meaningfully on received signals. CIC is reported in nats.

Counterfactual Decoding Effect (CDE) is a new metric developed for scrutinising the effect of the token on the receiver's next action kinematically; it measures the same receiver-side dependence directly in the action space. Let $\mu(o, m)$ denote the mean of the receiver's drive-action distribution $\pi^{\text{drive}}(\cdot \mid o, m)$, where o is the receiver's local observation and m is the tuple of tokens it receives from neighbours. At each recorded state the scene is held fixed (the sender's policy, action, and kinematics unchanged) and the received tokens are replaced by a counterfactual draw m'. CDE is the expected Euclidean shift in the mean action:

$$\mathrm{CDE} = \mathbb{E}_{s,\,m'\sim\mathrm{Unif}(\mathcal{M})}\left\|\mu(o_s,m') - \mu(o_s,m)\right\|_2, \qquad (18)$$

taken over recorded states s for which $m' \neq m$. Because μ is the distribution mean, no action-sampling noise enters, and the shift is attributable to decoding alone. We report the norm across the two action dimensions and, separately, the per-dimension magnitude of the longitudinal (acceleration) and lateral (steering) controls, in action units.

CDE is complementary to CIC. Both isolate the receiver's response to the token, but they measure it on different scales. CIC quantifies the influence as an information-theoretic divergence: it reports how much the message reduces uncertainty about the action, without a physical scale or a direction. CDE instead holds the whole scene fixed and reads the change directly in the action space, so it is expressed in the units of the control the agent applies and resolves which action dimension moves. In short, CIC asks whether the token affects the next action of the receiver, whereas CDE asks how much, and along which axis, the action itself shifts.

5) Intervention battery: A non-zero ρ shows the token and the driver's type are correlated. However, it does not tell us whether the token is genuinely coupled to behaviour or whether receivers act on it: the correlation could even be incidental, since if cautious agents happen to emit token 1 for kinematic reasons unrelated to any communicative act, ρ would be elevated even though the token plays no signalling role. To test the relationship between the token and the kinematics it is meant to govern, and whether receivers exploit it, we apply six interventions (Table III), each evaluated on the bimodal condition over 200 episodes.

Baseline. Everything works as trained: agents drive, emit tokens, and observe neighbours normally. *Hide.* Receivers are blind to the signal token; they can still see how neighbours drive, but the symbol is hidden. *Randomise.* A random draw replaces the token each agent emits, independent of type, so ρ is expected to be approximately 0 by construction under this condition. *Permute.* Token labels are scrambled with a fixed code – e.g., under $\pi = \{0\to2, 1\to0, 2\to1\}$, an agent emitting token 0 (intending conservative) triggers the assertive execution profile, while receivers still observe token 0 in the signal window; the label reaches receivers intact but its physical meaning is wrong. *Desync.* The token an agent emits is left untouched and stays visible to neighbours, but the execution profile it should trigger is re-drawn at random every step, independently of the token; neighbours see a genuine token history, but the kinematics behind it are pure noise. Desync and Permute both decouple the token from its kinematic consequence, but differ in whether the decoupling is stable: Permute applies a fixed scramble consistently, so a systematic (but wrong) token-kinematic relationship remains learnable, whereas Desync destroys that relationship entirely at every step. Greater degradation under Desync than Permute therefore indicates that receivers rely on the step-to-step kinematic image of the token rather than the token label alone. *Hide_kin.* Receivers can see the signal token but the driving behaviour of neighbours is hidden. *Hide_all.* Receivers see neither the token nor the driving behaviour of any neighbour.

All statistical tests use the Wilcoxon signed-rank test on paired seed-level differences from baseline, Holm–Bonferroni corrected across the six interventions within each metric; the

TABLE III: Intervention battery. Token and kinematics columns describe the state of each channel as seen by receivers or applied by the environment. **0** denotes zeroing; mismatched/random/step denotes kinematic execution decoupled from the emitted token.

Intervention	Token	Kinematics	What it tests
Baseline	intact	intact	Unmodified baseline.
Hide	**0**	intact	Is the discrete token m informationally necessary, or do kinematics alone suffice for receivers?
Randomise	$\mathrm{Unif}(\mathcal{M})$	intact	Does the type-signal convention carry load, or only the structural existence of the token channel?
Permute	intact	mismatched	Does the receiver rely on kinematic consistency with the token, or does it read the token symbolically?
Desync	intact	random/step	If the kinematic image is per-step noise but the token survives intact, does receiver behaviour degrade?
Hide_kin	intact	**0**	Is the kinematic channel necessary? (Direct complement of Hide.)
Hide_all	**0**	**0**	What is the total information value of the neighbour channel?

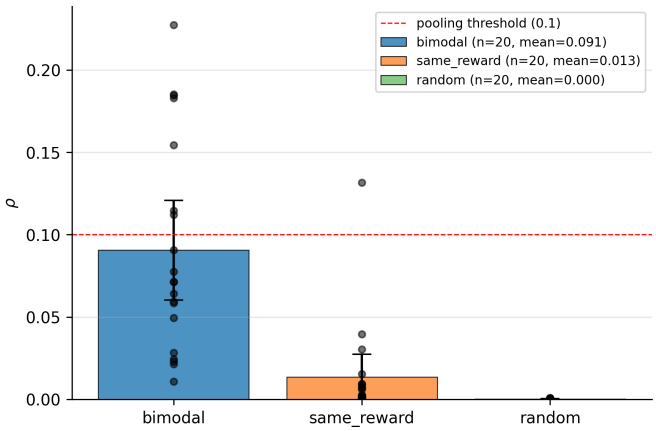


Fig. 2: Separation coefficient ρ by condition. Bars are means over 20 seeds with 95% seed-level confidence intervals; points are individual seeds.

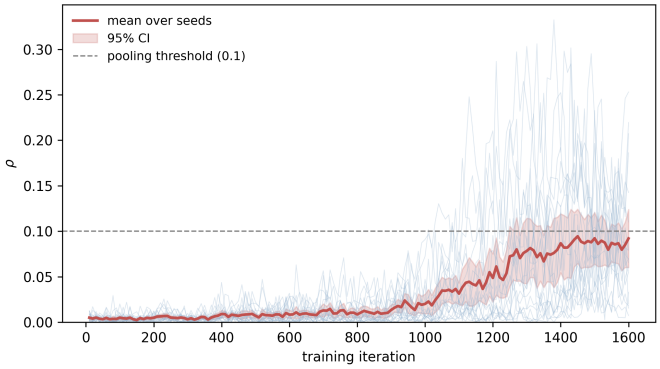


Fig. 3: ρ over training for the bimodal condition. Faint lines are the 20 individual seeds; the solid line is the mean, and the shaded band is the 95% seed-level confidence interval at each logged iteration; the dashed line marks the pooling threshold $\rho = 0.1$.

no-signal-versus-bimodal comparison uses a two-sided Mann–Whitney U test, since no-signal is an independently trained population.

IV. Results

The previous section set out the experimental conditions, the evaluation metrics, and the intervention battery this section draws on throughout. What follows reports what they showed: we begin with the separation coefficient, since it answers RQ1 and RQ2 directly, then turn to the intervention results and the receiver-side decoding measures.

The distribution of the separation coefficient across the three signalling conditions shown in Figure 2 answers both research questions. RQ1 asks whether type-contingent signalling emerges at all. In the bimodal condition, the token carried a small amount of type information, mean $\rho = 0.091$ of a maximum of 1.0. Moreover, it was statistically higher than both the same-reward and the random-signal controls. RQ1 was therefore answered: weak, type-associated signalling emerged, above the controls but far short of separation. On the operational cut used here ($\rho < 0.1$) the mean sat just inside the pooling range, but the seeds straddled it: 7 of the 20 exceeded 0.1, and the highest reached 0.227. The population was better described as weakly pooled on average than as uniformly pooled. We use the term throughout to describe the measured distribution, not to claim that the population reached a pooling equilibrium.

A single end-of-training value cannot distinguish an emergence that never formed from one that formed and then collapsed back to pooling. We therefore tracked ρ over training (Figure 3). Averaged over seeds, ρ rose early and then settled onto a stable plateau near its final value of about 0.09, without decaying back towards zero. Consequently, the weak type-signal association was a stable end state of training rather than a transient, and ρ stayed low throughout.

RQ2 asks whether the emergence is due to type-dependent reward heterogeneity rather than the signalling architecture itself. With the architecture and signalling head held fixed and only the reward difference between the types removed, ρ collapsed to near 0 (≈ 0.013). RQ2 was thus answered: emergent signalling was contingent on that reward heterogeneity, and disappeared once it was removed.

A low ρ does not, on its own, reveal how senders were emitting the token. Both types might be emitting more or less at random, or each might be settling firmly on one token while the two simply settle on the same one. Speaker Consistency

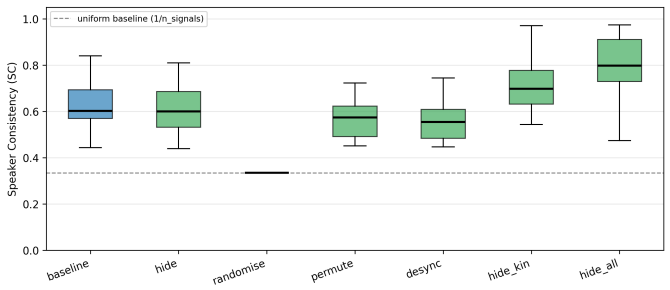


Fig. 4: Speaker Consistency by intervention, bimodal family. The dashed line marks the chance level $1/|\mathcal{M}| = 0.33$.

tells these apart, asking how strongly a single type sticks to its favoured token regardless of what the other type does. Across the bimodal seeds, each type held to its preferred token about 62% of the time, far above the 33% (select one out of three tokens) that random emission would give (Figure 4), so the choices were deliberate rather than noisy. Placed next to $\rho \approx 0.09$, the picture is pooling with consistency: every type commits to a token, but both commit to much the same one.

Read the other way round, the posterior $p(\theta \mid m)$ in Figure 5 gives the probability that a type is implied by a token. Each token is only weakly diagnostic: token 0 implied the cautious type about two-thirds of the time, while the others sat closer to the chance value of 0.5. This confirms weak, type-associated signalling rather than a type-revealing code. Because seeds settle on different conventions, averaging the posterior across seeds dampens the per-seed structure, so this figure understates the separation present within any single run.

Although it carries so little type information, the signalling token still has a meaningful effect on the system. The token and the kinematic channel carried complementary loads. Table IV shows that manipulating the token from its kinematic consequence in Permute, Randomise, and Desync raised the collision rate by 65% to 86% relative to baseline. Suppressing the neighbour kinematics (Hide_kin) produced the opposite signature: collisions were statistically unchanged (38.5% to 45.5%, corrected p=.866), while mean return fell by roughly two-thirds (201.4 to 66.4, corrected p<.001).

Interestingly, hiding the token from receivers while leaving neighbour driving intact (Hide) left safety essentially unchanged (collision rate 38.5% to 40.5%, corrected p=.659).

Figure 6 shows a population trained with no token at all crashed at 34.9% against the communicating population's 38.5%, a difference not detectable at twenty seeds per condition. As noted in Section III-H1, this no-signal condition is not a clean removal of the channel alone, and the comparison should be read with that in mind. Safety therefore does not depend on the token being present, or even on a binding existing in the first place. What raises the collision rate is specifically corrupting an established binding, so that the token and the behaviour it accompanies disagree. The presence or absence of either on its own does not.

A 65-to-86% rise in crashes when the token is scrambled could mean two different things: the agents genuinely rely on the token, or they are simply fragile to any disturbance at all. To tell these apart, we ran the same token interventions on two control families: the random-signal and the same-reward.

As shown in Figure 7, scrambling the token hurt safety a great deal in the bimodal, less in the same-reward, and not at all in the random-signal, where none of the token-corruption effects were significant. The agents that never learned to use the token were unharmed when it broke. The bimodal penalty is therefore genuine reliance on the token, not a general sensitivity to perturbation.

The interventions above alter the token only from the sender's perspective. To identify decoding from downstream behaviour, we evaluated the CIC and CDE.

According to Figure 8, in the bimodal condition the mean CDE shifted by 0.126 when the received token was scrambled, roughly four times that of the other two conditions. The shift was predominantly longitudinal (acceleration 0.099 against steering 0.043).

CIC gave the same ordering (Figure 9): 0.053 nats in the bimodal condition, about 4.8% of channel capacity (three tokens; $\ln 3 \approx 1.099$) and roughly seven times the other two controls (≈ 0.008). Together, they indicate that the token was decoded by the receiver. However, this is only a slight nudge to how the receiver actually acted.

Whether a seed developed more type information or stronger decoding did not translate into better driving. Across the twenty seeds, neither ρ nor CIC was significantly correlated with the collision rate or the mean return (Figure 10). The information the token carries and the strength with which receivers decode it are therefore decoupled from task performance, consistent with a channel whose safety load rests on the token-behaviour binding rather than on how much the symbol reveals.

In summary, across twenty seeds per condition, the emergent token carried little information about hidden type yet its integrity was worth a 65-to-86% rise in the collision rate. The load-bearing element was the binding between the token and the driving it commits to, not the receiver's reading of the symbol. Hiding the token from receivers left safety untouched at a return cost under 3% (corrected p=.040), whereas breaking the token-kinematic binding by any means raised collisions by 65% to 86%. The receiver-only counterfactual nonetheless confirmed that receivers do decode the token as a soft, speed-oriented cue decoupled from type separation. The kinematic channel carried the complementary efficiency load.

V. Discussion

In the previous section, ρ sat low, inside the range of the pooling regime. This does not mean that no signalling emerged; it means only that the token carries little information about a driver's hidden type. Which leaves a sharper question: did the agents signal at all, and if so, in what sense? Our answer, and the thread that runs through this section, is that weak type-associated signalling did emerge, in a sense the field's standard metrics are not built to detect: senders commit to a token pattern deliberately, receivers act on the token they observe, and the population comes to rely on the resulting token-behaviour coupling, enough that corrupting it raises

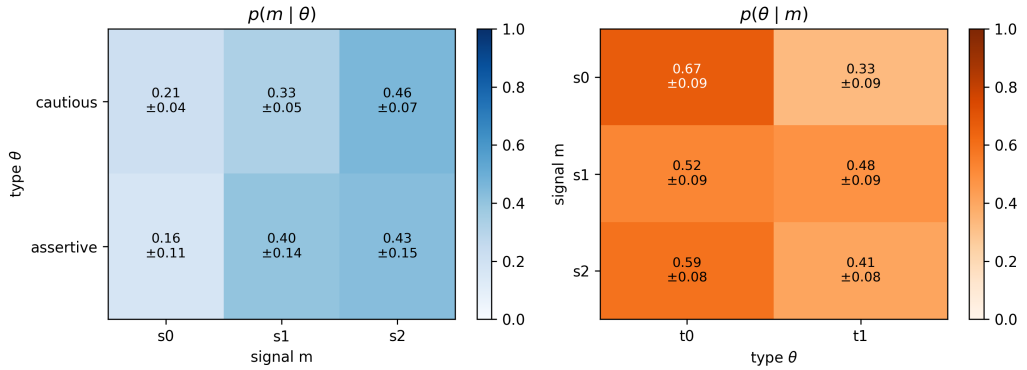


Fig. 5: Signal use in the bimodal condition, mean over 20 seeds. Left: $p(m \mid \theta)$, how each type distributes its tokens. Right: $p(\theta \mid m)$, the type implied by a received token. Error terms are 95% seed-level confidence intervals per cell.

TABLE IV: Evaluation-time interventions on the bimodal family (N=6, 20 seeds). Point values are seed-level means. Δ columns give the paired seed-level difference from baseline with a 95% t-CI. p is a two-sided Wilcoxon signed-rank test over the 20 paired seed-level differences, Holm–Bonferroni corrected across the six interventions separately within each metric.

	Collision rate			Return		
Intervention	%	Δ (%)	p	mean	Δ	p
Baseline	38.5	—	—	201.4	—	—
Hide	40.5	+2.1	.659	195.7	−5.8	.040
Permute	63.6	+25.1	.001	130.4	−71.0	<.001
Randomise	71.0	+32.5	<.001	110.3	−91.1	<.001
Desync	71.7	+33.2	<.001	110.5	−90.9	<.001
Hide_kin	45.5	+7.0	.866	66.4	−135.0	<.001
Hide_all	36.0	−2.5	.866	80.5	−120.9	<.001

collisions substantially. We read this as a convention of *use*, a shared regularity in how the token is emitted and read, rather than a convention of *meaning about type*, which is exactly what did not emerge.

Highway driving is a coordination-dominant, mixed-motive game. In Lewis's terms [12, p. 24], it is a decision-making situation involving two or more agents whose interests mostly coincide and who can coordinate on at least two different equilibrium outcomes. Concretely, there are many mutually compatible ways to interleave traffic, such as who leads a merge and who holds a lane; drivers only need to settle on one of them, since almost any consistent choice works, while a mismatch ends in a collision. It is worth noting that such a game is not pure coordination: coordinating still leaves who yields, who takes the gap, who leads the merge – the yielder loses time, the one who goes gains – so drivers agree they must coordinate but disagree which equilibrium they prefer. That is a mixed-motive game in the sense of Schelling [38, p. 89]. Highway driving is thus coordination-dominant but not pure coordination, mixed-motive at the edges.

What a message is worth in such a game depends on what one takes a signal to be for, and here game theory offers two quite different accounts. In the first account, a signal reveals something private about the sender. This is the logic of costly signalling [11] and of cheap talk [10], and it is what the separation coefficient ρ measures. Our channel is not cheap talk in the strict sense: emitting a token selects an execution profile, so it changes the sender's own kinematics, and therefore its payoff, whether or not any neighbour ever reads it. That places it closer to the costly-signalling side of the pair, and it is the reason the binding, rather than the symbol, turns out to be what matters. Judged this way, the agents barely communicate at all, since the token carries less than a tenth of a bit about a driver's hidden type. In the second account a signal need reveal nothing. Its only job is to let everyone settle on the same arbitrary convention out of the many that would serve equally well, in the sense of [12, pp. 16-24] and [38, pp. 54-58].

A coordination problem is not solely solved by revealing private information; it is solved by everyone landing on the same rule. For that reason our results read more naturally on the side of the second account, even though the metrics the field reaches for by default assume the first. The token says almost nothing about a driver's hidden type, yet the whole population organises its driving around it. It is not a failed code, it is what a convention looks like when the game rewards agreement over disclosure.

It helps to be precise about what kind of convention this is. A token can carry two very different conventions. The first is a *convention of meaning* about type, where a token comes to stand for cautious or assertive so that reading it tells you who the sender is. This is what ρ scores, and it is exactly what did not emerge. The second is a *convention of use*: a shared, self-consistent regularity in how the population emits the token and how receivers respond to it, whatever it reveals about type. It is this second convention that emerged: senders commit to a token pattern deliberately (Speaker Consistency well above chance), and receivers act on the token they observe

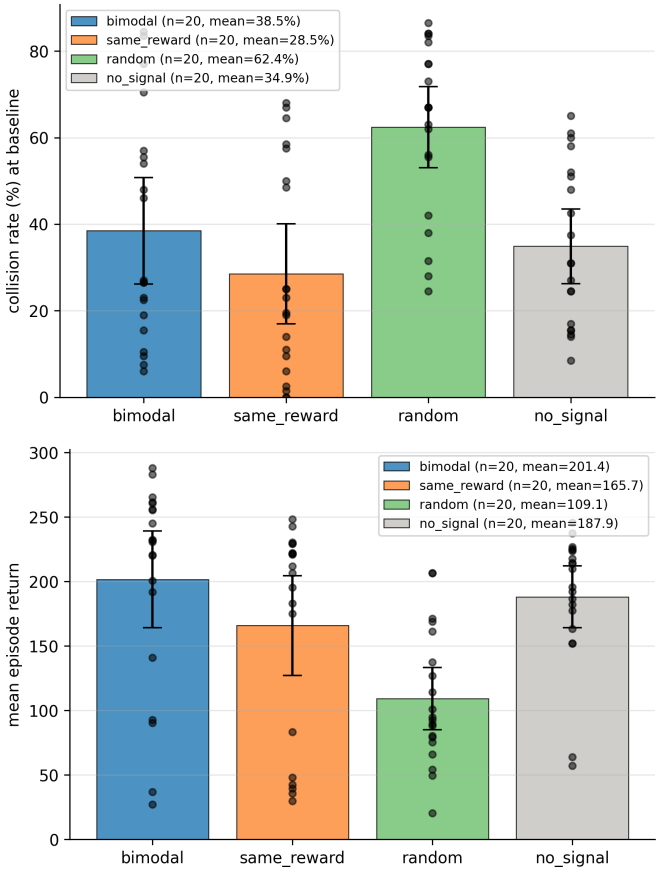


Fig. 6: Collision rate and return by condition under the unmodified baseline. Top: the collision rate. Bottom: the mean episode return. Bars are means over the 20 seeds of each condition, error bars are 95% seed-level confidence intervals, and points are individual seeds.

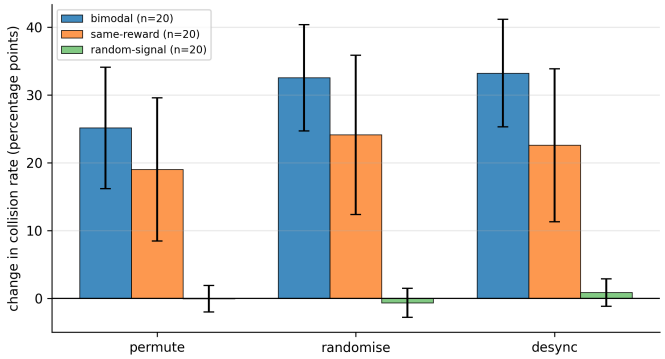


Fig. 7: Change in collision rate under the three token-corrupting interventions, by training family. Bars are means over the 20 seeds of each family and error bars are 95% seed-level confidence intervals on the paired difference from that family's own baseline. The penalty tracks whether a family had learned to use the token: large in the bimodal, smaller in the same-reward, and absent in the random-signal, whose agents never bound the token to behaviour.

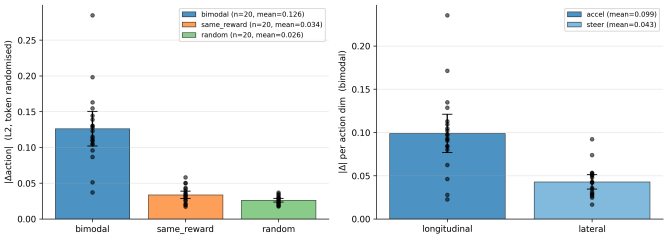


Fig. 8: Counterfactual Decoding Effect across conditions. Left: change in the receiver's mean action when the received token is altered with the scene frozen, by condition. Right: per-dimension change in the bimodal family. Bars are means over the 20 seeds, error bars are 95% seed-level confidence intervals, and points are individual seeds.

(non-zero decoding in every one of the twenty seeds). The population comes to rely on the resulting token-to-behaviour coupling, enough that corrupting it raises collisions by 65% to 86%. What emerged is a habit of use that coordinates driving, not a code that reveals type. We encourage readers to review example (1) Ship Flag Hoists and (2) The Trucker and The Driver (titles added by the authors) in [12, pp. 125-127, 129-130] for illustration of convention of use.

That the channel is a convention rather than a code shows first in how the token is used from one run to the next. Figure 5 shows that, on average, both types place most of their weight on token 2 (assertive kinematic envelope) though the preference is weak. The per-seed breakdown in Figure 11 tells the fuller story: individual seeds settle on different token conventions, and among the well-performing seeds several distinct choices reach comparable return. Different seeds assign the tokens differently – cautious agents commit to token 0 in one seed and to token 1 in another – yet both drive equally well. Relabelling the tokens in one seed's solution and it behaves like another seed's. No single labelling is privileged because the task never implicitly specifies which integer should mean which type, only that whichever mapping a population settles on stays consistent. The sharply higher collision rate and lower mean return under Permute, Desync, and Randomise are what a mismatch in a coordination problem looks like: sender and receiver are operating on different conventions, and the outcome degrades accordingly.

This also settles what it means to speak of a signal's meaning. As the experimenter we label each token with the driving profile it triggers, token 2 with the assertive envelope for instance, but that label records only the grounding we built in, the mechanical consequence of emitting the symbol; it is not the token's communicative meaning. What a token signifies as a signal is fixed by how the population comes to use it, and that is a matter of convention, not of our labelling. This is Lewis's point about signalling systems [12, pp. 130-132]: a signal has no intrinsic meaning; its meaning just is the sender-receiver regularity the population adopts. We supply the vocabulary and its physical consequences and the agents supply the semantics.

If the particular labelling is arbitrary, why should the token matter at all? The answer is that its value never lay in what it reveals. Across twenty seeds the token carries almost nothing

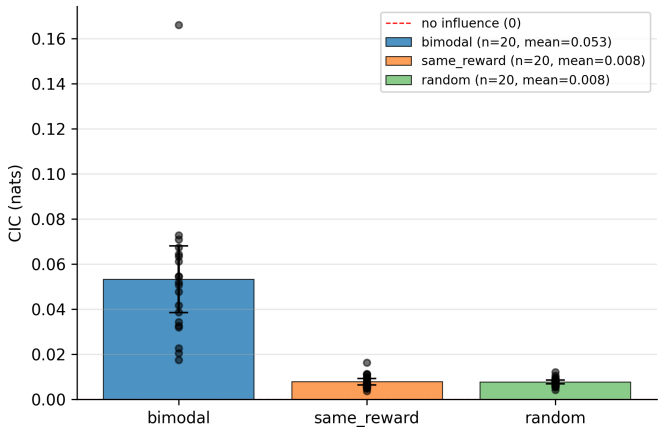


Fig. 9: Causal Influence of Communication by condition (nats). Bars are means over the 20 seeds, error bars are 95% seed-level confidence intervals, and points are individual seeds.

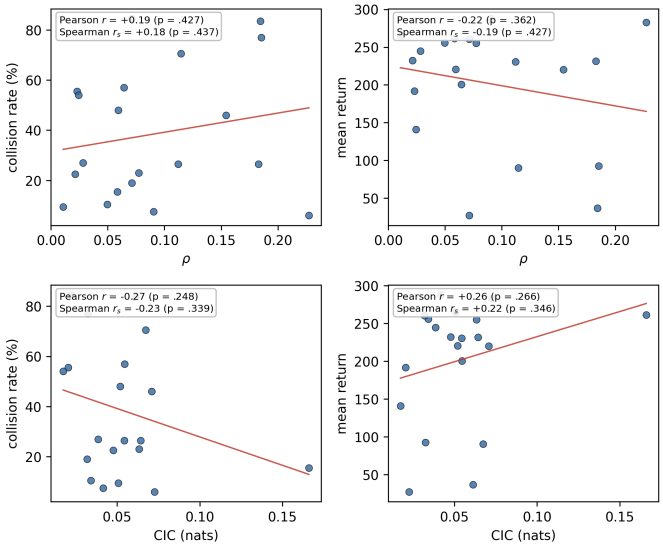


Fig. 10: Seed-level relationships between the signalling metrics (ρ, CIC) and task outcomes (collision rate, mean return) across 20 bimodal seeds. Each point is one seed and the line is an ordinary least-squares fit. Each panel is annotated with the Pearson r and the Spearman r_s over the 20 seeds, with the associated p-value. All four relationships are weak and none is significant.

about a driver's hidden type ($\rho \approx 0.09$, with not one seed reaching the separating threshold), yet the population organises its driving around it. Corrupting the token-to-behaviour binding lifts the collision rate by 65% to 86%. A signal can thus be informationally near-silent and still be load-bearing.

Why the population pooled at all is a question about learning rather than about payoffs. Nothing in the design forced it. With three tokens available and only two types to tell apart, a separating code was within reach and would have cost nothing to represent; the channel had more capacity than the task required. The agents settled on using the tokens in much the same way whatever their type anyway.

[39] studies this question for Lewis signalling games played by agents who learn by reinforcement rather than by computing equilibria. In the smallest such game the learning process converges on a signalling system with probability one. That guarantee is fragile, though: enlarge the game, and a run can converge on a partial pooling outcome with positive probability. A population can then settle into pooling that no one would have chosen deliberately, with the early and essentially arbitrary reinforcements deciding which outcome it reaches. Pooling on this account is an attractor of the learning process, not a property of the game being played.

The setting in such a game is different from ours – a Lewis signalling game with two states, two signals, two acts, using simple Roth-Erev reinforcement learning [40, 41] – so we do not claim the theorems transfer. What carries over is the shape of the explanation, and it fits our empirical results. ρ rises early and then flattens (Figure 3) rather than climbing towards separation, even though we specifically added a signal-entropy bonus to prevent that (Section III-F). Consequently, it looks like a learning outcome rather than slow progress. Individual seeds land on different token conventions and stay there (Figure 11). Read this way, the near-empty token is not evidence that the task had no use for an informative signal; it is what a population of learners can settle into even when an informative signal was there to be found.

If the token is doing coordinating work, one might expect removing it to hurt. Figure 6 points out it does not. A population that never had the channel crashes at 34.9% against the communicating population's 38.5%, and returns 187.9 against 201.4. Neither difference is detectable at twenty seeds per condition. The channel looks, in this sense, optional to adopt: the population reaches much the same place with it, without it, or ignoring it.

What is not optional is the cost of disrupting the relationship once it exists. The same token whose presence is dispensable becomes costly to corrupt, as breaking the token-to-behaviour binding lifts collisions by 65% to 86% over baseline. We read this as consistent with the convention of use described above, where emitting a token and driving the matching envelope tend to go together and receivers appear to have learned to read one from the other. On that reading, corrupting the binding does not remove information, since the token carried almost none. Instead it injects incoherence: a driver whose token and manoeuvre disagree breaks a regularity other drivers appear to expect. That mismatch is plausibly, though not directly shown, what the others crash into. A casual analogy is a traffic light. When a light loses power and goes dark, drivers fall back to a safe default, treating it as an all-way stop and creeping through, so traffic slows but almost nobody crashes. When a light instead malfunctions and shows green in every direction at once, it is actively misinforming, and confident drivers pull into each other. A missing light is safe; a misinformative light is a disaster.

A signal can be a convention and be informative at the same time. The signalling token here ended up carrying so little about type that it is worth asking why. The interventions locate the answer in where the useful information already lives. Strip the token but leave the kinematics and receivers lose only

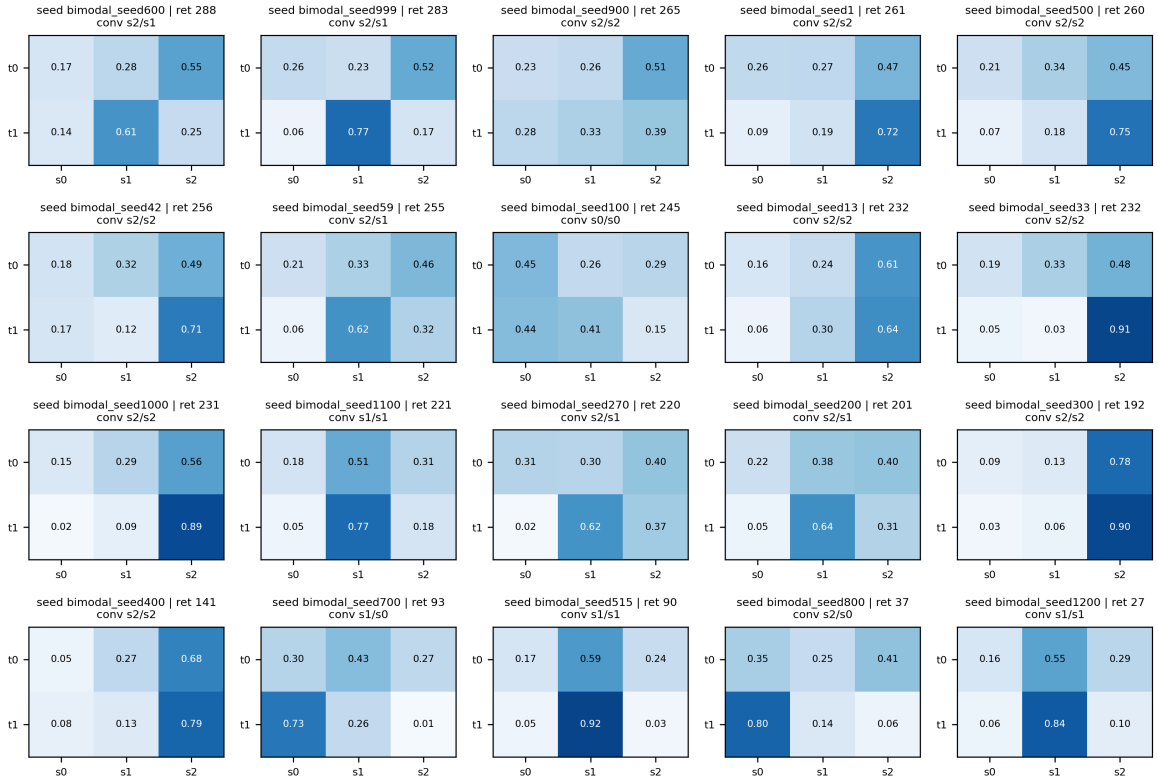


Fig. 11: Signal heatmap by seed in the bimodal condition, sorted by descending mean return.

a few percent; strip the kinematics but leave the token and return collapses to about a third of baseline. What a receiver needs is written in the motion, not in the symbol. Because the channel is grounded, a driver's type already shows in how it accelerates and holds its lane, so the token has little left to add.

Redundancy removes the incentive: the objective pays only for driving outcomes, not for informative symbols, and once the kinematics reveal type there is nothing to pay for encoding it again. This is close to the classic condition under which cheap talk stays uninformative [10], though the parallel is loose, since our token is not costless: it commits the sender to an execution profile. The redundancy argument does not need that result to stand; it needs only that the kinematics already carry type and that nothing in the reward pays for encoding it a second time. In our experiments, agents still performed the task without communication, showing that a good policy was reachable without it. The listener therefore ignored the sender's messages, and consequently the speaker had no incentive to produce meaningful signals [18]. On this reading the empty token is not a failure of the setup but the expected outcome of learning a signal no one is paid to make informative.

We cannot fully separate could-not from need-not, whether the agents were unable to find an informative code or simply had no reason to; but with type already legible in the kinematics, redundancy is the more persuasive reading. Our design deliberately gives no reward that would steer agents towards any particular way of emitting signals. It is possible that with such a reward the token could become more informative about an agent's type, but we leave that to future work.

A convention only communicates if the receivers actually read it, and they do. Freezing the scene and swapping only the received token moves a receiver's next action about four times as much as in the controls (Figure 8). The information-theoretic version of the same test puts the influence at roughly seven times control levels (Figure 9). Both measures are non-zero in every one of the twenty seeds, and they agree closely with each other across seeds. The decoding is soft rather than dramatic, but it is real and it is everywhere. Paired with the Speaker Consistency score, it supplies both halves of [17]'s operational definition of communication, positive signalling and positive listening. The strength of that decoding is also decoupled from ρ and from Speaker Consistency: receivers condition on the token about equally whether or not it carries type information. This is the convention of use at work, positive signalling paired with positive listening, and it holds even where the revelation metric reads essentially zero.

Taken at face value, the passive numbers are unremarkable. ρ sits in the pooling regime, and CIC, though reliably above control levels, is small in absolute magnitude. A reader who stopped at the scoreboard would hastily conclude that these small numbers are not the result of communication. Yet the same channel, when its binding is broken, costs a 65-to-86% rise in collisions. The interventions disagree about how much the token matters, and they are right: the dependence is real and large, and the scores simply do not see it.

The reason is that the existing metrics are instruments of the information-theoretic tradition. Mutual information asks

how much a signal tells us about a hidden state, but it is silent about the load carried by a signal that reveals nothing yet binds an action others depend on. We mean binding here in the mechanical sense that emitting a token forces a matching execution profile. A convention of this kind is invisible to a passive audit and shows itself only under active perturbation. [17] and [18] recommend exactly this: to certify that a channel is used, one must intervene on it and watch behaviour move. Our methodology and interventions offer a different perspective on emergent communication in multi-agent systems generally, and in multi-agent autonomous driving in particular: communication can occur not by being informative, but by establishing a behavioural convention.

These findings carry a practical reading for connected and autonomous driving. The lesson for a vehicle-to-vehicle protocol is that the quantity worth protecting is the binding between message and manoeuvre, not the mere presence of messages or their informational richness. A vehicle can broadcast an accurate self-report but act inconsistently with it. Because existing information metrics score only the honesty of what is said about type, not the coupling between message and behaviour, they can certify a channel that is in fact behaviourally unreliable. Evaluation of connected AVs should therefore test the message-to-manoeuvre binding directly, not just the informativeness of the messages. This is in line with Vehicle-to-Everything (V2X) misbehaviour-detection and intent-recognition research that already looks for inconsistency between what a vehicle signals and how it drives. We advance this as an implication rather than a demonstration, since our evidence comes from a single simulated environment, but the direction it points is clear.

Several limitations bound these claims. First, the evidence comes from a single simulated environment at small scale, six agents on one straight highway with no background traffic. Whether the same regime holds under a different road layout, more agents, or scripted background traffic remains an open question. Second, the population mix is also held fixed at an even split. We ran only this balanced bimodal composition rather than a sweep over type proportions, so it is unknown how the emergent regime depends on the balance of cautious and assertive drivers, and whether a skewed population would push it toward or away from separation. The no-signal comparison also confounds two factors, since that condition removes the token and fixes the execution profile at neutral in the same step; separating them would need a fifth condition in which agents keep the three profiles but select among them without emitting anything a neighbour can read. Fourth, the experiments are solely simulated under one scenario, driving on the highway; further experiments should be conducted under other scenarios such as merging, driving at a roundabout, or an intersection. Finally, the mechanism identified here rests on kinematic cost asymmetries built into a simplified simulator; the dynamics of the simulation environment are not validated against real driving data, and it is unclear whether the same claims would survive real vehicle dynamics, sensor noise, and imperfect perception.

VI. Conclusion

This paper asked whether communication emerges among cooperative reinforcement-learning drivers given a grounded discrete channel, and in what sense. It does, but not in the form standard metrics register. Across seeds the token carried almost no information about a driver's hidden type, yet corrupting the token-behaviour binding substantially raised the collision rate, while a population trained without the channel showed no detectable difference in collision rate or return. Receivers decoded the token regardless of how little it revealed: the Counterfactual Decoding Effect introduced here and the Causal Influence of Communication agreed closely, both non-zero in every seed tested. What emerged is better described as a convention-of-use than a type-revealing code: informationally near-silent, but load-bearing in behaviour.

The methodological point follows from these results rather than a general claim about metrics: information-theoretic scores measure revelation, and here they registered very little, while breaking the token-behaviour binding cost a substantial rise in collisions on the same runs. This is reason to test the coupling between what an agent signals and how it then drives, rather than read informativeness scores alone – a design principle for cooperative multi-agent systems generally, where a signal that changes the sender's own behaviour should have that coupling tested directly. For connected autonomous vehicles, the practical implication is that what needs protecting is not a message's content but its binding to the manoeuvre it selects; we offer this as an implication of the result, not a demonstrated engineering claim. These claims rest on a single small-scale simulated highway with an even, binary type split, a confounded no-signal control, and simulator dynamics not validated against real driving; broader traffic scenarios, larger populations, and real-world validation are left to future work.

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APPENDIX A
NOTATION

Table V lists the notation used throughout the paper. Structural constants are repeated in Table VI.

TABLE V: Notation used throughout.

Symbol	Meaning
Θ, θ	Set of latent driver types; a single type. $\lvert\Theta\rvert = 2$.
$\mathcal{M}$, m	Set of discrete tokens; a single emitted token. $\lvert\mathcal{M}\rvert = 3$.
N	Number of controlled agents ($= 6$).
K	Number of observed neighbours per agent ($= 5$).
W	History window of past tokens observed ($= 5$).
L	Number of lanes ($= 3$).
o	Receiver's local observation (kinematics of neighbours).
$p(\theta, m)$	Empirical joint frequency of type θ emitting token m.
$p(\theta)$, $p(m)$	Marginal type and token frequencies.
$p(m \mid \theta)$	Emission distribution: token given type.
$p(\theta \mid m)$	Posterior: type given token.
$I(\Theta; \mathcal{M})$	Mutual information between type and emitted token (bits).
ρ	Separation coefficient, $I(\Theta;\mathcal{M})/\log\lvert\Theta\rvert \in [0,1]$.
SC	Speaker Consistency, $\frac{1}{\lvert\Theta\rvert}\sum_\theta \max_m p(m \mid \theta)$.
CIC	Causal Influence of Communication (nats).
CDE	Counterfactual Decoding Effect (action units).
π^{drive}	Receiver's drive-action policy.
$\mu(o, m)$	Mean of the receiver's drive-action distribution.
D_{KL}	Kullback–Leibler divergence.

APPENDIX B
HYPERPARAMETERS

Table VI lists the training hyperparameters used for every condition reported in the paper.

TABLE VI: Hyperparameters.

Parameter	Value
Learning rate	3×10^{-4}
Rollout steps T	512
PPO epochs per iteration	4
Minibatch size	128
PPO clip ratio ϵ	0.2
GAE λ	0.95
Discount γ	0.99
Value coef c_v	0.5
Signal entropy start c_m^0	0.08
Signal entropy end c_m^{end}	0.005
Anneal iters T_{anneal}	1200
Aux coef c_{aux}	0.5
Gradient clip norm	0.5
Total iterations	1600
TypeEncoder hidden dimension d	64
DeepSets output dimension d_{agg}	64
Signal/response hidden dimension	64
Critic hidden dimension	128

APPENDIX C
HARDWARE, SOFTWARE, AND SIMULATION ENVIRONMENT CONFIGURATION

Table VII lists the hardware and software used to run all experiments, and Table VIII lists the simulation environment configuration.

TABLE VII: Hardware and software configuration.

Component	Specification
Processor	Apple M4 Max (14 cores)
Memory	36 GB unified
Machine	MacBook Pro M4 Max
Operating system	macOS 26.5.2 (build 25F84)
Accelerator	Apple Metal (MPS); CUDA not used
Python	3.11.15
PyTorch	2.11.0
`highway-env`	1.10.2
Gymnasium	1.3.0

TABLE VIII: Simulation environment configuration.

Component	Specification
Controlled agents N	6
Background (IDM) vehicles	0
Lanes L	3
Episode duration	40 s
Control frequency	5 Hz
Observation	Multi-agent kinematics, 6 vehicles per agent (ego plus $K = 5$ neighbours), relative, normalised
Action space	Continuous longitudinal + lateral, clipped to $[-1, 1]$
Reward speed range	20–30 m/s
Seeds per condition	20

APPENDIX D
STATISTICAL METHODOLOGY

All intervals and error bars are 95% confidence intervals. For a metric summarised across the twenty seeds of a condition (ρ, Speaker Consistency, CIC, CDE, collision rate, mean return), the interval is a Student-t interval over the seed-level values; each seed contributes one value, itself an average over its 200 evaluation episodes. For the training curve in Figure 3, we plot the mean of ρ across seeds at each logged iteration with a 95% confidence interval. Paired intervention-versus-baseline tests use the Wilcoxon signed-rank test with a Holm–Bonferroni correction. Cross-seed relationships use both Pearson and Spearman correlations. The *no-signal* versus *bimodal* baseline comparison uses the two-sided Mann–Whitney U test rather than a paired test, since the *no-signal* condition is an independently trained population. Exact and adjusted p-values for every test reported in the paper are given below.

TABLE IX: Paired intervention-versus-baseline tests on the *bimodal* family ($n = 20$ seeds; baseline collision rate 38.5%, baseline return 201.4). p is the two-sided Wilcoxon signed-rank test on the 20 paired seed-level differences between that intervention and the baseline; p_{Holm} is the Holm–Bonferroni correction applied across the six interventions, separately within each metric.

	Collision rate		Return	
Intervention	p	p_{Holm}	p	p_{Holm}
Hide	0.220	0.659	0.0400	0.0400
Permute	2.91×10^{-4}	1.17×10^{-3}	3.62×10^{-5}	1.45×10^{-4}
Randomise	8.83×10^{-5}	5.30×10^{-4}	3.81×10^{-6}	1.91×10^{-5}
Desync	8.83×10^{-5}	5.30×10^{-4}	1.91×10^{-6}	1.14×10^{-5}
Hide_kin	0.550	0.866	1.05×10^{-4}	2.10×10^{-4}
Hide_all	0.433	0.866	4.77×10^{-5}	1.45×10^{-4}

TABLE X: Paired intervention-versus-baseline tests on the *same-reward* family ($n = 20$ seeds; baseline collision rate 28.5%, baseline return 165.7).

	Collision rate		Return	
Intervention	p	p_{Holm}	p	p_{Holm}
Hide	0.324	0.324	0.784	0.784
Permute	0.001	0.003	1.91×10^{-5}	7.63×10^{-5}
Randomise	6.34×10^{-4}	0.003	2.67×10^{-5}	8.01×10^{-5}
Desync	6.69×10^{-4}	0.003	6.29×10^{-5}	1.26×10^{-4}
Hide_kin	4.49×10^{-4}	0.002	1.91×10^{-6}	1.14×10^{-5}
Hide_all	1.91×10^{-6}	1.14×10^{-5}	1.91×10^{-6}	1.14×10^{-5}

TABLE XI: Paired intervention-versus-baseline tests on the *random-signal* family ($n = 20$ seeds; baseline collision rate 62.4%, baseline return 109.1).

	Collision rate		Return	
Intervention	p	p_{Holm}	p	p_{Holm}
Hide	0.926	1.000	0.985	1.000
Permute	0.911	1.000	0.729	1.000
Randomise	0.456	1.000	0.674	1.000
Desync	0.295	1.000	0.388	1.000
Hide_kin	0.277	1.000	0.002	0.008
Hide_all	0.627	1.000	0.001	0.006

TABLE XII: *No-signal* versus *bimodal* baseline, unpaired ($n = 20$ seeds each). Two-sided Mann–Whitney U test on the seed-level values.

Metric	No-signal	Bimodal	U	p
Collision rate	34.9%	38.5%	203.0	0.946
Mean return	187.9	201.4	265.0	0.081

TABLE XIII: Cross-seed correlations on the *bimodal* family ($n = 20$ seeds).

	Pearson		Spearman	
Relationship	r	p	r_s	p
ρ vs collision rate	$+0.188$	0.427	$+0.184$	0.437
ρ vs mean return	-0.215	0.362	-0.188	0.427
CIC vs collision rate	-0.271	0.248	-0.226	0.339
CIC vs mean return	$+0.261$	0.266	$+0.223$	0.346
CDE vs CIC	$+0.896$	9.63×10^{-8}	$+0.756$	1.14×10^{-4}

DECLARATIONS

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